

H4 Integration rules A level only

Name: _____

Class: _____

Date: _____

Time: **308 minutes**

Marks: **257 marks**

Comments:

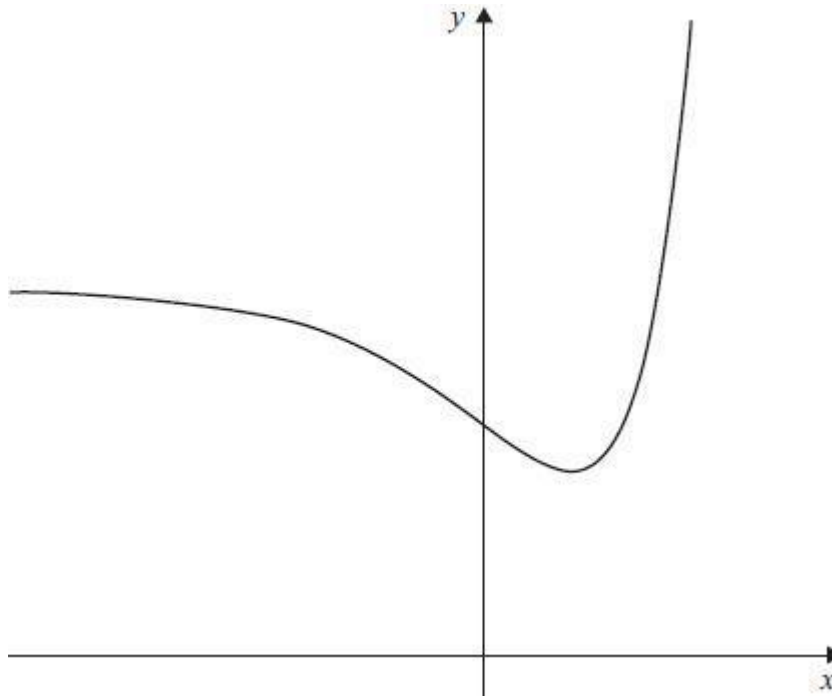
EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Q1.

A function f has domain $\mathbb{R}$ and range $\{y \in \mathbb{R} : y \geq e\}$

The graph of $y = f(x)$ is shown.



The gradient of the curve at the point (x, y) is given by $\frac{dy}{dx} = (x - 1)e^x$

Find an expression for $f(x)$.

Fully justify your answer.

©2025 Exam Papers Practice. All rights reserved.

(Total 8 marks)

Q2.

(a) Prove the identity $\frac{\sin 2x}{1 + \tan^2 x} \equiv 2\sin x \cos^3 x$

(3)

(b) Hence find $\int \frac{4 \sin 4\theta}{1 + \tan^2 2\theta} d\theta$

(6)

(Total 9 marks)

Q3.

- (a) Given that $u = 2^x$, write down an expression for $\frac{du}{dx}$ (1)

- (b) Find the exact value of $\int_0^1 2^x \sqrt{3+2^x} \, dx$

Fully justify your answer.

(6)

(Total 7 marks)

Q4.

$\int_1^2 x^3 \ln(2x) \, dx$ can be written in the form $p \ln 2 + q$, where p and q are rational numbers.

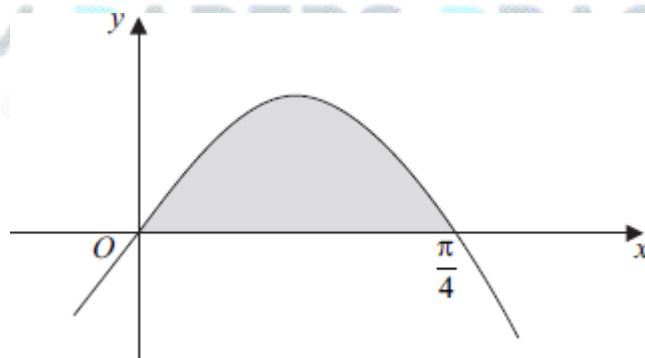
Find p and q .

(Total 5 marks)

Q5.

A curve has equation $y = 4x \cos 2x$.

- (a) Find an exact equation of the tangent to the curve at the point on the curve where $x = \frac{\pi}{4}$. (5)
- (b) The region shaded on the diagram below is bounded by the curve $y = 4x \cos 2x$ and the x -axis from $x = 0$ to $x = \frac{\pi}{4}$.



By using integration by parts, find the exact value of the area of the shaded region.

(5)

(Total 10 marks)

Q6.

- (a) Show that

$$\int_0^{\ln 2} e^{1-2x} dx = \frac{3}{8}e$$

(4)

- (b) Use the substitution $u = \tan x$ to find the exact value of

$$\int_0^{\frac{\pi}{4}} \sec^4 x \sqrt{\tan x} dx$$

(8)

(Total 12 marks)

Q7.

(a) Find $\int x\sqrt{x^2 + 3} dx$.

(2)

- (b) Solve the differential equation

$$\frac{dy}{dx} = \frac{x\sqrt{x^2 + 3}}{e^{2y}}$$

given that $y = 0$ when $x = 1$. Give your answer in the form $y = f(x)$.

(7)

(Total 9 marks)

Q8.

- (a) (i) By writing $\ln x$ as $(\ln x) \times 1$, use integration by parts to find $\int \ln x dx$.

(4)

(ii) Find $\int (\ln x)^2 dx$.

(4)

- (b) Use the substitution $u = \sqrt{x}$ to find the exact value of $\int_1^4 \frac{1}{x + \sqrt{x}} dx$.

(7)

(Total 15 marks)

Q9.

(a) Find $\int t \cos\left(\frac{\pi}{4}t\right) dt$.

(4)

- (b) The platform of a theme park ride oscillates vertically. For the first 75 seconds of the

ride,

$$\frac{dx}{dt} = \frac{t \cos\left(\frac{\pi}{4}t\right)}{32x}$$

where x metres is the height of the platform above the ground after time t seconds.

At $t = 0$, the height of the platform above the ground is 4 metres.

Find the height of the platform after 45 seconds, giving your answer to the nearest centimetre.

(6)

(Total 10 marks)

Q10.

- (a) Given that $y = 4x^3 - 6x + 1$, find $\frac{dy}{dx}$.

(1)

- (b) Hence find $\int_2^3 \frac{2x^2 - 1}{4x^3 - 6x + 1} dx$, giving your answer in the form $p \ln q$, where p and q are rational numbers.

(5)

(Total 6 marks)

Q11.

- (a) Given that $x = \frac{1}{\sin \theta}$, use the quotient rule to show that $\frac{dx}{d\theta} = -\operatorname{cosec} \theta \cot \theta$.

(3)

- (b) Use the substitution $x = \operatorname{cosec} \theta$ to find $\int_{\sqrt{2}}^2 \frac{1}{x^2 \sqrt{x^2 - 1}} dx$, giving your answer to three significant figures.

(9)

(Total 12 marks)

Q12.

Solve the differential equation

$$\frac{dy}{dx} = y^2 x \sin 3x$$

given that $y = 1$ when $x = \frac{\pi}{6}$. Give your answer in the form $y = \frac{9}{f(x)}$.

(Total 9 marks)

Q13.

Use the substitution $u = x^4 + 2$ to find the value of $\int_0^1 \frac{x^7}{(x^4 + 2)^2} dx$, giving your answer in the form $p \ln q + r$, where p , q and r are rational numbers.

(Total 6 marks)

Q14.

- (a) A water tank has a height of 2 metres. The depth of the water in the tank is h metres at time t minutes after water begins to enter the tank. The rate at which the depth of the water in the tank increases is proportional to the difference between the height of the tank and the depth of the water.

Write down a differential equation in the variables h and t and a positive constant k .

(You are not required to solve your differential equation.)

(3)

- (b) (i) Another water tank is filling in such a way that t minutes after the water is turned on, the depth of the water, x metres, increases according to the differential equation

$$\frac{dx}{dt} = \frac{1}{15x\sqrt{2x-1}}$$

The depth of the water is 1 metre when the water is first turned on.

Solve this differential equation to find t as a function of x .

(8)

- (ii) Calculate the time taken for the depth of the water in the tank to reach 2 metres, giving your answer to the nearest 0.1 of a minute.

(1)

(Total 12 marks)

Q15.

- (a) Find $\int \frac{1}{3+2x} dx$.

(2)

- (b) By using integration by parts, find $\int x \sin \frac{x}{2} dx$.

(4)

(Total 6 marks)

Q16.

Use the substitution $u = 1 + 2 \tan x$ to find

$$\int \frac{1}{(1+2 \tan x)^2 \cos^2 x} dx$$

(Total 5 marks)

Q17.

- (a) Using integration by parts, find $\int x \sin(2x - 1) dx$.

(5)

- (b) Use the substitution $u = 2x - 1$ to find $\int \frac{x^2}{2x - 1} dx$, giving your answer in terms of x .

(6)

(Total 11 marks)

Q18.

- (a) Given that $y = \frac{4x}{4x - 3}$, use the quotient rule to show that $\frac{dy}{dx} = \frac{k}{(4x - 3)^2}$, where k is an integer.

(2)

- (b) (i) Given that $y = x \ln(4x - 3)$, find $\frac{dy}{dx}$.

(3)

- (ii) Find an equation of the tangent to the curve $y = x \ln(4x - 3)$ at the point where $x = 1$.

(3)

- (c) (i) Use the substitution $u = 4x - 3$ to find $\int \frac{4x}{4x - 3} dx$, giving your answer in terms of x .

(4)

- (ii) By using integration by parts, or otherwise, find $\int \ln(4x - 3) dx$.

(4)

(Total 16 marks)

Q19.

- (a) Use integration by parts to find $\int (t - 1) \ln t dt$.

(4)

- (b) Use the substitution $t = 2x + 1$ to show that $\int 4x \ln(2x + 1) dx$ can be written as $\int (t - 1) \ln t dt$.

(3)

- (c) Hence find the exact value of $\int_0^1 4x \ln(2x + 1) dx$.

(3)

(Total 10 marks)

Q20.

- (a) (i) Given that $y = 2x^2 - 8x + 3$, find $\frac{dy}{dx}$.

(1)

- (ii) Hence, or otherwise, find

$$\int_4^6 \frac{x - 2}{2x^2 - 8x + 3} dx$$

giving your answer in the form $k \ln 3$, where k is a rational number.

(4)

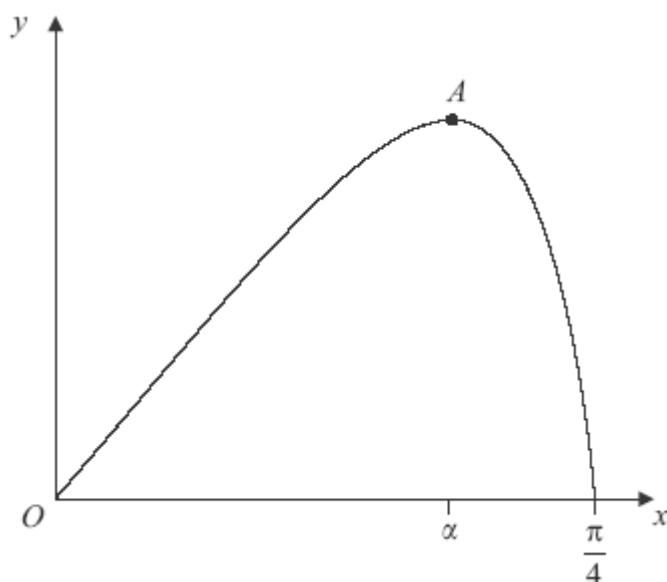
- (b) Use the substitution $u = 3x - 1$ to find , giving your answer in terms of x .

(4)

(Total 9 marks)

Q21.

A curve is defined for $0 \leq x \leq \frac{\pi}{4}$ by the equation $y = x \cos 2x$, and is sketched below.



- (a) Find $\frac{dy}{dx}$. (2)
- (b) The point A , where $x = \alpha$, on the curve is a stationary point.
- (i) Show that $1 - 2\alpha \tan 2\alpha = 0$. (2)
- (ii) Show that $0.4 < \alpha < 0.5$. (2)
- (iii) Show that the equation $1 - 2x \tan 2x = 0$ can be rearranged to become $x = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x}\right)$. (1)
- (iv) Use the iteration $x_{n+1} = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x_n}\right)$ with $x_1 = 0.4$ to find x_3 , giving your answer to two significant figures. (2)
- (c) Use integration by parts to find $\int_0^{0.5} x \cos 2x dx$, giving your answer to three significant figures. (5)

(Total 14 marks)

Q22.

- (a) Given that $y = \frac{\sin \theta}{\cos \theta}$, use the quotient rule to show that $\frac{dy}{d\theta} = \sec^2 \theta$. (3)
- (b) Given that $x = \sin \theta$, show that $\frac{x}{\sqrt{1-x^2}} = \tan \theta$. (2)
- (c) Use the substitution $x = \sin \theta$ to find $\int \frac{1}{(1-x^2)^{\frac{3}{2}}} dx$, giving your answer in terms of x . (5)
- (Total 10 marks)

Q23.

- (a) Use integration by parts to find $\int x e^{5x} dx$. (4)
- (b) (i) Use the substitution $u = \sqrt{x}$ to show that
- $$\int \frac{1}{\sqrt{x}(1+\sqrt{x})} dx = \int \frac{2}{1+u} du$$
- (2)
- (ii) Find the exact value of $\int_1^9 \frac{1}{\sqrt{x}(1+\sqrt{x})} dx$ (3)
- (Total 9 marks)
- ©2025 Exam Papers Practice. All rights reserved.

Q24.

- (a) (i) Given that $f(x) = x^4 + 2x$, find $f'(x)$. (1)
- (ii) Hence, or otherwise, find $\int \frac{2x^3 + 1}{x^4 + 2x} dx$. (2)
- (b) (i) Use the substitution $u = 2x + 1$ to show that

$$\int x\sqrt{2x+1} dx = \frac{1}{4} \int \left(u^{\frac{3}{2}} - u^{\frac{1}{2}} \right) du$$

(3)

- (ii) Hence show that $\int_0^4 x\sqrt{2x+1} \, dx = 19.9$ correct to three significant figures.

(4)

(Total 10 marks)

Q25.

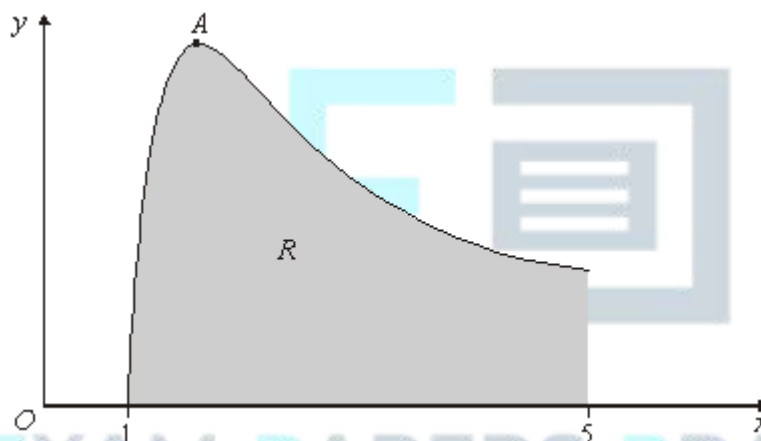
- (a) Given that $y = x^{-2} \ln x$, show that $\frac{dy}{dx} = \frac{1 - 2 \ln x}{x^3}$.

(4)

- (b) Using integration by parts, find $\int x^{-2} \ln x \, dx$.

(4)

- (c) The sketch shows the graph of $y = x^{-2} \ln x$.



- (i) Using the answer to part (a), find, in terms of e, the x-coordinate of the stationary point A.

(2)

- (ii) The region R is bounded by the curve, the x-axis and the line $x = 5$. Using your answer to part (b), show that the area of R is

$$\frac{1}{5} (4 - \ln 5)$$

(3)

(Total 13 marks)

Q26.

- (a) Find $\frac{dy}{dx}$ when $y = (3x - 1)^{10}$.

(2)

- (b) Use the substitution $u = 2x + 1$ to find $\int x(2x + 1)^8 dx$, giving your answer in terms of x .

(4)

(Total 6 marks)

Q27.

- (a) Find $\int e^{4x} dx$.

(1)

- (b) Use integration by parts to find $\int e^{4x}(2x + 1) dx$.

(3)

- (c) By using the substitution $u = 1 + \ln x$, or otherwise, find $\int \frac{1 + \ln x}{x} dx$.

(4)

(Total 8 marks)



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.